

COMM114: Generative AI - VAE (cont.)

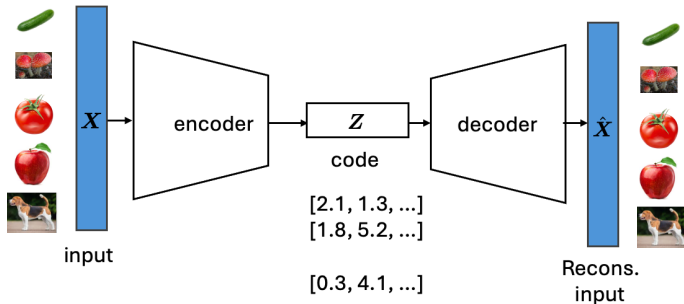
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Revisiting AEs (cont.)

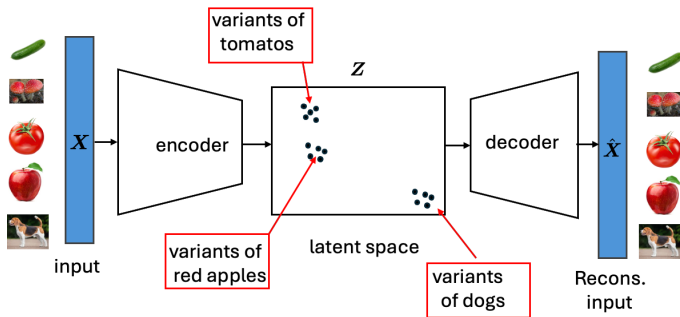
Reason for AE not being a generative model.

- When the AE is trained, only the reconstruction error between the input $\mathbf{X}$ and the output $\hat{\mathbf{X}}$ is considered. No additional function on the code $\mathbf{Z}$ is being enforced.
- Therefore, input samples of similar semantics (e.g., different images of tomatos) do **NOT** necessarily lead to the property that their latent codes are close to each other.



VAE

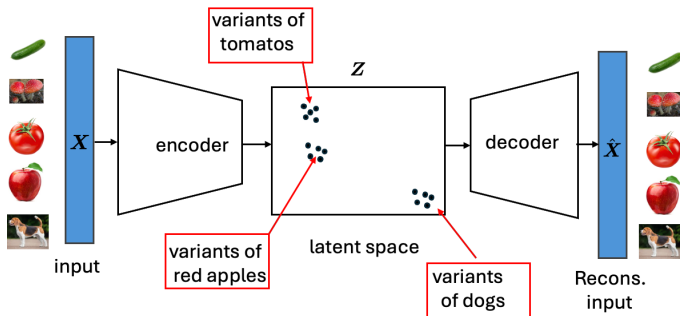
- The variational autoencoder (VAE), instead of learning a code, learns a '**latent space**'. The latent space is characterized by a parametric distribution.



- The semantic relationships between the data samples are captured by their latent samples in the latent space.

Demonstration of semantic relationships

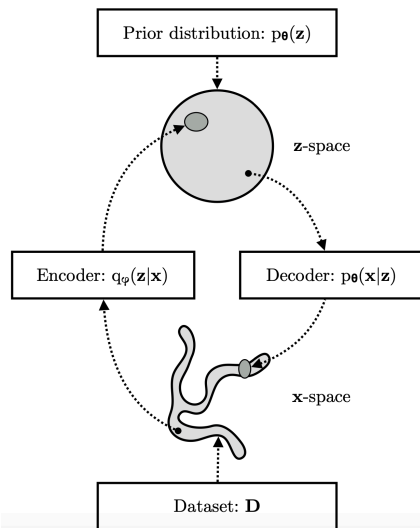
- Latent samples for variants of tomatos are close to each other.
- Latent samples for variants of red apples are close to each other.
- Latent samples for red apples and those for tomatos are relatively close to each other.



VAE

- Different from that of AE, a VAE learns stochastic mappings between an observed $\mathbf{x}$ -space and a latent $\mathbf{z}$ space.
- The generative model learns a joint distribution $p_{\theta}(\mathbf{x}, \mathbf{z}) = p_{\theta}(\mathbf{z})p_{\theta}(\mathbf{x}|\mathbf{z})$. The stochastic decoder learns the conditional probability $p_{\theta}(\mathbf{x}|\mathbf{z})$.
- $p_{\theta}(\mathbf{z})$ is the prior distribution of the latent random variable $\mathbf{z}$.
- The stochastic encoder learns an approximation $q_{\varphi}(\mathbf{z}|\mathbf{x})$ of the true but intractable posterior $p_{\theta}(\mathbf{z}|\mathbf{x})$ of the generative model.
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Remark: $q_{\varphi}(\mathbf{z}|\mathbf{x})$ is referred to as the approximate posterior.

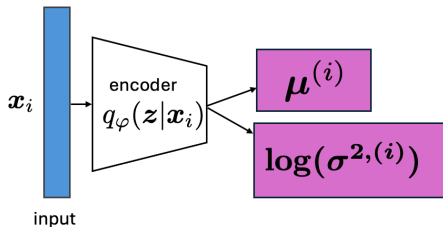


VAE

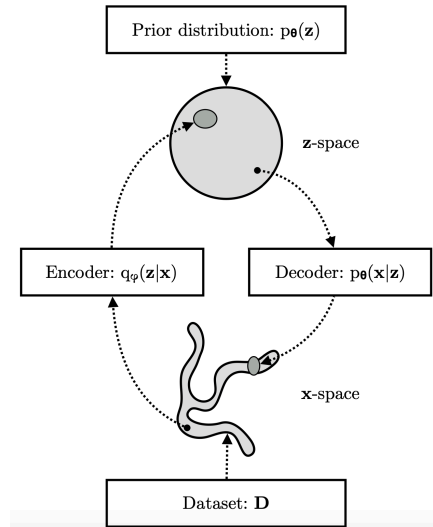
Procedure for simplifying the training process

- The prior distribution is set to be $p_{\theta}(\mathbf{z}) = \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$
- The approximate posterior $q_{\varphi}(\mathbf{z}|\mathbf{x})$ is set to be a multivariate Gaussian with a diagonal covariance matrix

$$q_{\varphi}(\mathbf{z}|\mathbf{x}_i) = \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}^{(i)}, \sigma^{2,(i)} \mathbf{I})$$



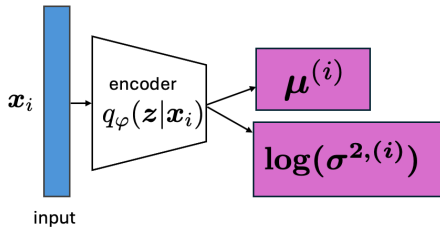
Remark: The vector $\boldsymbol{\mu}^{(i)}$ and $\log(\sigma^{2,(i)})$ are functions of φ and $\mathbf{x}_i$.



VAE

Exercise:

- Why does the encoder in a VAE output $\log(\sigma^{2,(i)})$ instead of $(\sigma^{2,(i)})$?



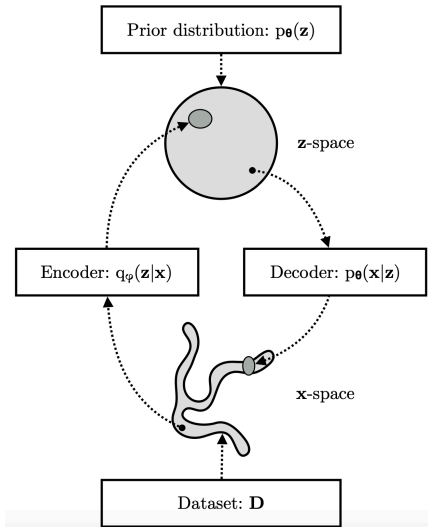
VAE

Procedure for simplifying the training process

- The conditional distribution $p_{\theta}(\mathbf{x}|\mathbf{z})$ is set to be either multivariate Gaussian in case of real valued input data or Bernoulli in case of binary input data.
- For the case of Gaussian distribution, $p_{\theta}(\mathbf{x}|\mathbf{z})$ takes the form of

$$p_{\theta}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}; f_{\theta}(\mathbf{z}), \mathbf{I}),$$

where $f_{\theta}(\mathbf{z})$ is a deep neural network.



Objective function in a VAE

The stochastic encoder (parameterized by φ) and the stochastic decoder (parameterized by θ) in a VAE are trained by maximizing a so-called **evidence lower bound (ELBO)** function:

$$\begin{aligned} & \max_{\theta, \varphi} \mathcal{L}(\theta, \varphi) \\ &= \max_{\theta, \varphi} \frac{1}{n} \sum_{i=1}^n [\mathcal{L}(\theta, \varphi; \mathbf{x}_i)] \\ &= \max_{\theta, \varphi} \frac{1}{n} \sum_{i=1}^n [-D_{KL}(q_{\varphi}(\mathbf{z}|\mathbf{x}_i) || p_{\theta}(\mathbf{z})) + \mathbb{E}_{q_{\varphi}(\mathbf{z}|\mathbf{x}_i)} [\log p_{\theta}(\mathbf{x}_i|\mathbf{z})]] \\ &= \max_{\theta, \varphi} \frac{1}{n} \sum_{i=1}^n \left[-D_{KL}(\mathcal{N}(\mathbf{z}; \boldsymbol{\mu}^{(i)}, \boldsymbol{\sigma}^{2, (i)} \mathbf{I}) || \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})) + \mathbb{E}_{q_{\varphi}(\mathbf{z}|\mathbf{x}_i)} [\log p_{\theta}(\mathbf{x}_i|\mathbf{z})] \right] \end{aligned}$$

Objective function in a VAE (cont.)

The term $\mathbb{E}_{q_{\varphi}(\mathbf{z}|\mathbf{x}_i)} [\log p_{\theta}(\mathbf{x}_i|\mathbf{z})]$ in the **ELBO** function can be approximated by:

$$\begin{aligned} & \max_{\theta, \varphi} \mathcal{L}(\theta, \varphi) \\ &= \max_{\theta, \varphi} \frac{1}{n} \sum_{i=1}^n \left[-D_{KL}(\mathcal{N}(\mathbf{z}; \boldsymbol{\mu}^{(i)}, \boldsymbol{\sigma}^{2, (i)} \mathbf{I}) || \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})) + \mathbb{E}_{q_{\varphi}(\mathbf{z}|\mathbf{x}_i)} [\log p_{\theta}(\mathbf{x}_i|\mathbf{z})] \right] \\ &\approx \min_{\theta, \varphi} \frac{1}{n} \sum_{i=1}^n \left[D_{KL}(\mathcal{N}(\mathbf{z}; \boldsymbol{\mu}^{(i)}, \boldsymbol{\sigma}^{2, (i)} \mathbf{I}) || \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})) - \frac{1}{L} \sum_{l=1}^L [\log p_{\theta}(\mathbf{x}_i|\mathbf{z}_{i,l})] \right] \quad (1) \end{aligned}$$

where

$$\mathbf{z}_{i,l} = \boldsymbol{\mu}^{(i)} + \boldsymbol{\sigma}^{(i)} \odot \boldsymbol{\epsilon}_l,$$

where $\boldsymbol{\epsilon}_l$ is a Gaussian noise drawn from the prior distribution $p_{\theta}(\mathbf{z})$,

$$\boldsymbol{\epsilon}_l \sim \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I}).$$

Remark: The expectation is approximated by a summation over L random samples $\{\mathbf{z}_{i,l}\}_{l=1}^L$. The notation $\odot$ denotes elementwise product.

Objective function in a VAE (cont.)

Exercise:

Suppose $\epsilon_l \sim \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$, and $\mathbf{z}_{i,l} = \boldsymbol{\mu}^{(i)} + \boldsymbol{\sigma}^{(i)} \odot \epsilon_l$.

- Is $\mathbf{z}_{i,l}$ Gaussian distributed?
- What are the mean and covariance of $\mathbf{z}_{i,l}$?

Objective function in a VAE (cont.)

Consider the case that $p_{\theta}(\mathbf{x}_i|\mathbf{z}_{i,l})$ in (1) is Gaussian distribution:

$$p_{\theta}(\mathbf{x}_i|\mathbf{z}_{i,l}) = \mathcal{N}(\mathbf{x}_i; f_{\theta}(\mathbf{z}_{i,l}), \mathbf{I})$$

The quantity $-\frac{1}{L} \sum_{l=1}^L \log p_{\theta}(\mathbf{x}_i|\mathbf{z}_{i,l})$ in (1) can be simplified to be the MSE loss:

$$\begin{aligned} & \arg \min_{\theta, \varphi} - \sum_{i=1}^n \frac{1}{L} \sum_{l=1}^L \log p_{\theta}(\mathbf{x}_i|\mathbf{z}_{i,l}) \\ &= \arg \min_{\theta, \varphi} - \frac{1}{L} \sum_{l=1}^L \log \left[\frac{1}{Z} e^{-\frac{1}{2}(\mathbf{x}_i - f_{\theta}(\mathbf{z}_{i,l}))^2} \right] \\ &= \arg \min_{\theta, \varphi} - \frac{1}{L} \sum_{l=1}^L \left[-\frac{1}{2}(\mathbf{x}_i - f_{\theta}(\mathbf{z}_{i,l}))^2 + \text{constant} \right] \\ &= \arg \min_{\theta, \varphi} \frac{1}{L} \sum_{l=1}^L [(\mathbf{x}_i - f_{\theta}(\mathbf{z}_{i,l}))^2] \end{aligned} \tag{2}$$

Objective function in a VAE (cont.)

The ELBO function by plugging (2) into (1) is given by

$$\arg \min_{\boldsymbol{\theta}, \boldsymbol{\varphi}} \frac{1}{n} \sum_{i=1}^n \left[\underbrace{D_{KL}(\mathcal{N}(\mathbf{z}; \boldsymbol{\mu}^{(i)}, \boldsymbol{\sigma}^{2,(i)} \mathbf{I}) || \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I}))}_{\text{1st term}} + \frac{1}{L} \sum_{l=1}^L \underbrace{(\mathbf{x}_i - f_{\boldsymbol{\theta}}(\mathbf{z}_{i,l}))^2}_{\text{2nd term}} \right] \quad (3)$$

- The 2nd term minimize the reconstruction error between each input sample $\mathbf{x}_i$ and its associated output $f_{\boldsymbol{\theta}}(\mathbf{z}_{i,l})$.
- The first term makes the approximate posterior $\mathcal{N}(\mathbf{z}; \boldsymbol{\mu}^{(i)}, \boldsymbol{\sigma}^{2,(i)} \mathbf{I})$ of the stochastic encoder as close as possible to the prior distribution $\mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$.

Objective function in a VAE (cont.)

Simplify the first term in (3):

$$\begin{aligned} & \arg \min_{\boldsymbol{\theta}, \boldsymbol{\varphi}} D_{KL}(\mathcal{N}(\mathbf{z}; \boldsymbol{\mu}^{(i)}, \boldsymbol{\sigma}^{2, (i)} \mathbf{I}) || \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})) \\ & \vdots \\ & = \arg \min_{\boldsymbol{\theta}, \boldsymbol{\varphi}} \frac{1}{2} \sum_{j=1}^k \left(1 + \log(\boldsymbol{\sigma}^{(i)}[j])^2 - (\boldsymbol{\mu}^{(i)}[j])^2 - (\boldsymbol{\sigma}^{(i)}[j])^2 \right) \end{aligned} \quad (4)$$

where $\boldsymbol{\sigma}^{(i)}[j]$ and $\boldsymbol{\mu}^{(i)}[j]$ are the j th element of the two vectors $\boldsymbol{\mu}^{(i)}$ and $\boldsymbol{\sigma}^{(i)}$.

Remark: The detailed derivation will be provided in the workshop.

Remark: k denotes the dimension of $\mathbf{z}$.

Final objective function in a VAE

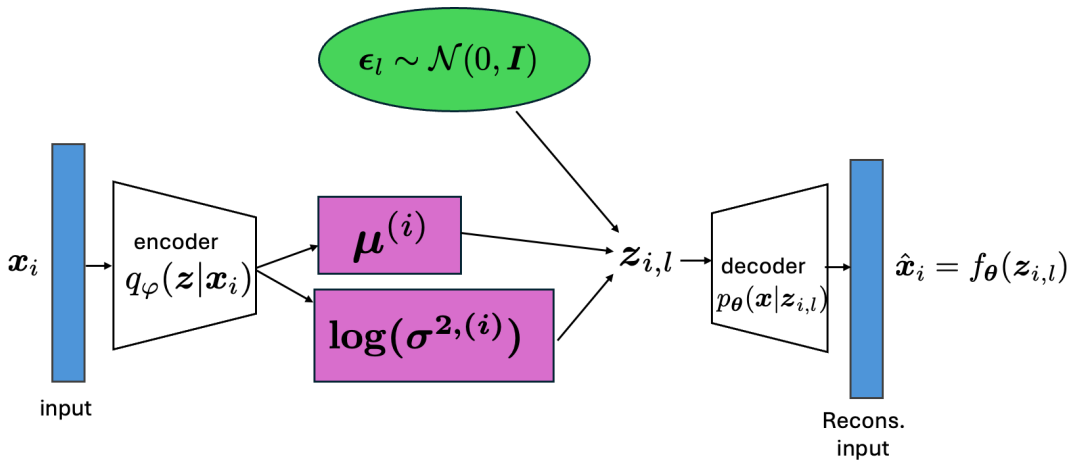
The final ELBO function is given by

$$\arg \min_{\boldsymbol{\theta}, \boldsymbol{\varphi}} \frac{1}{n} \sum_{i=1}^n \left[\underbrace{\frac{1}{2} \sum_{j=1}^k \left(1 + \log(\boldsymbol{\sigma}^{(i)}[j])^2 - (\boldsymbol{\mu}^{(i)}[j])^2 - (\boldsymbol{\sigma}^{(i)}[j])^2 \right)}_{\text{1st term}} + \frac{1}{L} \sum_{l=1}^L \underbrace{(\mathbf{x}_i - f_{\boldsymbol{\theta}}(\mathbf{z}_{i,l}))^2}_{\text{2nd term}} \right], \quad (5)$$

where $\mathbf{z}_{i,l} = \boldsymbol{\mu}^{(i)} + \boldsymbol{\sigma}^{(i)} \odot \boldsymbol{\epsilon}_l$, and $\boldsymbol{\epsilon}_l \sim \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$.

- The two vectors $\boldsymbol{\mu}^{(i)}$ and $\boldsymbol{\sigma}^{(i)}$ are functions of $\boldsymbol{\varphi}$.
- As a result, the 2nd term is a function of both $\boldsymbol{\varphi}$ and $\boldsymbol{\theta}$.

Diagram when training a VAE



- For each input sample $\mathbf{x}_i$, one can obtain a number of different reconstructions $f_\theta(\mathbf{z}_{i,l})$ depending on different Gaussian noises ϵ_l , $l = 1, \dots, L$.

Training a VAE

- Input: Data samples $\{\mathbf{x}_i\}_{i=1}^n$
- Specify two DNN architectures for both the encoder and the decoder.
- Apply stochastic gradient descent (SGD) to minimize the ELBO in (5). Each time, only a subset of training samples are considered when updating the model parameters.

References

- Diederik P. Kingma and Max Welling, "An Introduction to Variational Autoencoders", <https://arxiv.org/abs/1906.02691>, 2019.
- [Youtube video by Umar Jamil](#)

Questions?